

Role of patient specific 3D printed models in patient confidence, understanding and satisfaction of care in Singapore

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ARTICLE INFO

Keywords:

Patient specific 3D models
Customised 3D models
Preoperative counselling
Improve understanding of patients' condition
Boost patients' confidence in the procedure

ABSTRACT

Introduction: Patient specific 3D models have been widely used for pre-op planning and intra-op guidance in orthopaedic surgery. These models however are not often used in pre-operative doctor-patient communication. This study evaluates the roles of customized 3D models in improving patient understanding, confidence, and satisfaction of patient care when they were used during preoperative counselling.

Materials and methods: A prospective survey was conducted on 33 orthopaedic trauma patients who were required to rate on a scale of 1–5, the effectiveness of patient specific 3D models in: 1) improving patient's understanding and, 2) helping patients cope with the condition, 3) boosting patients' confidence in the treatment and 4) in the surgeon; and on a scale of 0–10, their overall satisfaction. Subgroup analysis was done to compare ratings of patients by age and by education levels.

Results: Over 90% patients rated agree or strongly agree on customised 3D models' effectiveness in improving understanding of injury and boosting confidence in treatments and surgeons. 87% patients agreed or strongly agreed that the models enhanced patient self-efficacy. No significant correlation was identified between age and patients' perceived effectiveness of customised 3D models in improving patient care. Ratings on four areas evaluated by pre-secondary and post-secondary groups were comparable. Post-secondary group had significantly higher satisfaction level than the pre-secondary group.

Conclusion: Customized 3D models help patients visualise complex pathology to facilitate patients' understanding of their condition and treatment, resulting in improved self-efficacy, confidence, and overall satisfaction. The use of patient specific 3D models in pre-operative counselling allows greater patient involvement therefore prompting patient-centred healthcare. Age does not influence patients' perceived effectiveness of customised 3D models in improving patient care. Patients with higher education level are likely to experience higher satisfaction level due to their willingness to take responsibility for their care.

1. Introduction

3D printing has gained increasing popularity in the field of orthopaedic surgery in recent years. This innovative technology has diverse applications in the surgery process, including producing personalised implants, model building for preoperative planning and providing intraoperative guidance.¹ Customised 3D models have proven to be effective in reducing operative time and cost-saving in surgery when they are used for preoperative planning or as jigs used intraoperatively.² These models have also been widely used in the training of physicians and students³ but have not to find widespread adoption in doctor-patient communications or patient education. Traditionally,

surgeons heavily rely on 2D scans such as CT and MRI imaging or standardised models which do not show structural anomalies to explain to patients their conditions and treatments. These 2D representations of injuries or anatomical abnormalities are not intended for patients who do not have relevant medical knowledge to understand the extend of injury therefore compromising on the quality of physician patient communication.⁴ Studies have also shown that using 3D models in doctor-patient communication allows greater patient engagement and performed better than 2D imaging and 3D reconstruction in educating patients on their condition and treatment plans in maxillofacial surgery, prostate and kidney cancer treatment, orthopaedic surgery, and other disciplines.^{5–7} Studies have shown that effective doctor-patient

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<https://doi.org/10.1016/j.jor.2024.02.010>

Received 5 February 2024; Accepted 9 February 2024

Available online 10 February 2024

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education has positive association with patient experience and clinical effectiveness, and patient safety.⁸

The Medical 3D Printing centre of a tertiary hospital has incorporated patient specific 3D models of complex injuries as part of the pre-operative planning. The models were created based on scans and produced in-house, as a complement to 2D images to aid surgeons in making optimal treatment plans. Examples of 3D models used include fractures of the femur, proximal humerus, orbital floor, and pelvis. In addition to preoperative planning, these models were employed by surgeons during preoperative consultation to better explain to patients the precise extent of injury, aims of surgery, post-op rehabilitation as well as recovery. There has been extensive research on the use of customised 3D models for surgery and for medical training but there is a scarcity of literature that explores the utilisation of patient specific models in patient education. This paper studied the roles of customized 3D models in enhancing patients' understanding of their condition, bolstering their confidence in prescribed treatments and healthcare providers, fostering self-efficacy in coping with the condition, and ultimately overall satisfaction with the care they received when these models were used during preoperative counselling in Singapore. This paper also explored how different patient characteristics can influence the effectiveness of customised 3D models in enhancing patients' care.

2. Methods

2.1. Study design

Patient specific 3D printed anatomical models were created based on CT (computer tomographic) scans and were subsequently used as an adjunct tool to educate the patients on their condition, the procedure planned, surgical risks and complications prior to the surgery. Critical anatomy and pathologies were featured in the form of colour variation to facilitate visualisation of key structures by patients (Fig. 1). A prospective survey was conducted on 33 patients to gather feedback on 3D printed models used during preoperative consultation. General information including age, race, highest education level, occupation, type of anatomical model used during preoperative counselling was collected. Patients were required to rate on a scale of 1–5, the effectiveness of 3D models in 1) improving patients' understanding of the condition, 2) helping patients to cope with the condition better 3) boosting patients' confidence in the treatment 4) increasing confidence in the surgeon providing the treatment. Patients were then asked to rate on a scale of 0–10 (0 being least satisfied and 10 being most satisfied) on the level of satisfaction of patient care they received as shown in Fig. 2. The survey also asked about patients' willingness to pay for the model and how much they were willing to pay.

2.2. Statistical analysis

A thorough analysis was then done using Microsoft version 2307 and



Fig. 1. Bimalleolar ankle fracture (Left). Scapula fracture (Right).

SPSS 26 to understand the perceived effectiveness of patient-specific 3D models in facilitating preoperative communication with patients. A bar chart was plotted to reflect different ratings in percentages for each aspect evaluated in survey. Subgroup analysis was conducted to explore potential correlations between patients' age and ratings for the four aspects evaluated in the survey, as well as between age and satisfaction level. Normality of each data set was checked with the Shapiro Wilk test. Spearman's correlation coefficient was calculated for non-parametric data and Pearson's correlation coefficient was calculated for parametric test. Patients were stratified into two groups based on their highest level of education which were pre-secondary education and post-secondary education. Pre-secondary education institutions include primary schools and secondary schools. Post-secondary education institutions include junior college, polytechnics, universities, and post-graduate education (masters and PhD). Mean ratings and satisfaction scores of patients with different level of education were compared using Mann-Whitney *U* test (for nonparametric data) or *T* test (for parametric data). *P* value less than 0.05 was considered significant.

3. Results

Over 87% of patients rated 4 and above (agree and strongly agree) for all aspects evaluated as shown in Fig. 3. One patient voted '1' (strongly disagree) on the effectiveness of customised 3D models in helping the patient to cope with his or her condition. More than half of the patients rated the maximum of 5 out of 5 for the effectiveness of patient specific 3D models in improving their understanding of the condition (70%), confidence in the treatment provided (61%) and confidence in the doctor treating them (64%) whereas 45% patients strongly agreed that the model helped them to cope with their condition. For overall satisfaction, 27% patients gave a rating of 10/10 and 94% patients rated 7 and above (Fig. 4).

Weak and negative correlations were identified between age and the effectiveness of 3D models in improving patients' understanding of their condition and helping them to cope with the condition (Spearman coefficient = -0.063 and -0.187 respectively) as shown in Table 1. On the other hand, patients' age was found to have a positive correlation with how well customised models can improve patients' confidence in the treatment provided (Spearman coefficient = $.100$). The correlations identified were statistically insignificant. There was no correlation between age and the effectiveness of the models' in improving patients' confidence in the doctor. Negative correlation was also noted between age and overall satisfaction level (Spearman coefficient = -0.247 , $p = 0.083$). Although the mean ratings by patients in the post-secondary education group were higher for each aspect evaluated in the survey compared to ratings by patients in the pre-secondary group, the differences in the mean ratings between the two groups were not significant. However, overall level satisfaction showed statistically significant differences between the two groups ($p = 0.019$). The mean overall level of satisfaction was 7.94 in pre-secondary school, and 9.06 in post-secondary group as shown in Table 2.

4. Discussion

3D printed patient specific instrumentations (PSIs) have been widely used across orthopaedic surgery for accurate implant and equipment placement especially in the presence of complex injury patterns in trauma cases. Clinically, these instruments have shown favourable outcomes with reduced operation time, decrease in intraoperative blood loss and less use of fluoroscopy.⁹ In the field of orthopaedic trauma, there are limited studies on the use of customised 3D models in preoperative counselling. Most of the existing studies focus on comparing 3D patient specific models with conventional 2D imaging^{10,11} yet no study has looked at patient characteristics that can possibly influence the effectiveness of these customised models in improving patient care which were discussed in our subgroup analysis.

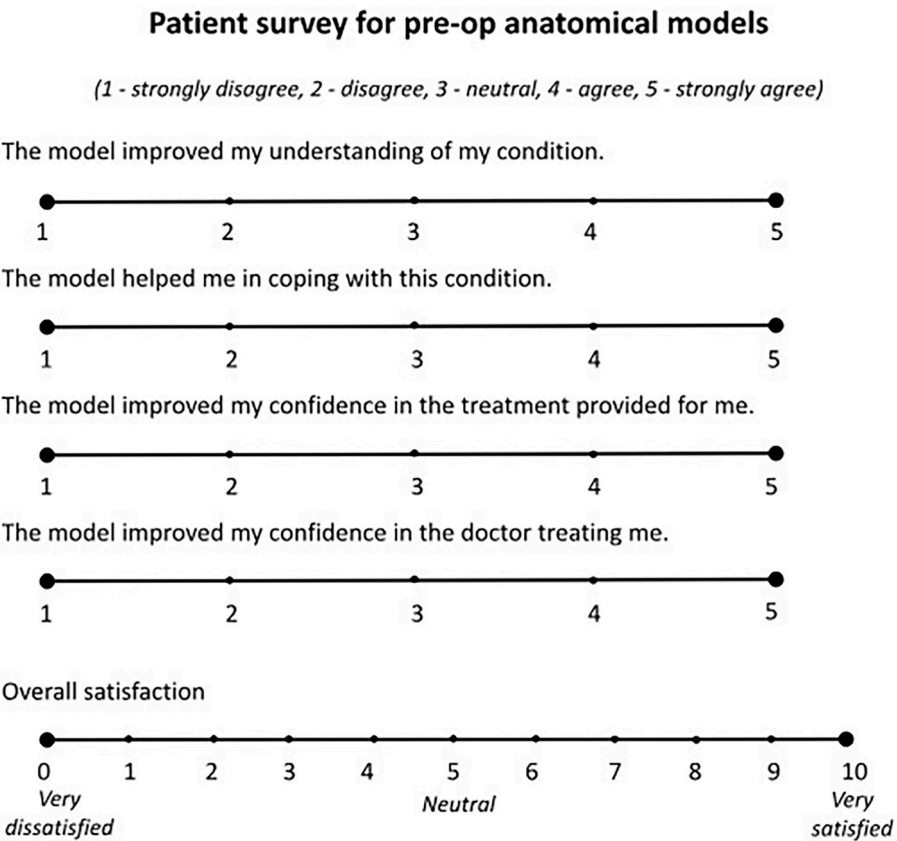


Fig. 2. Survey sample for preoperative patient specific 3D models.

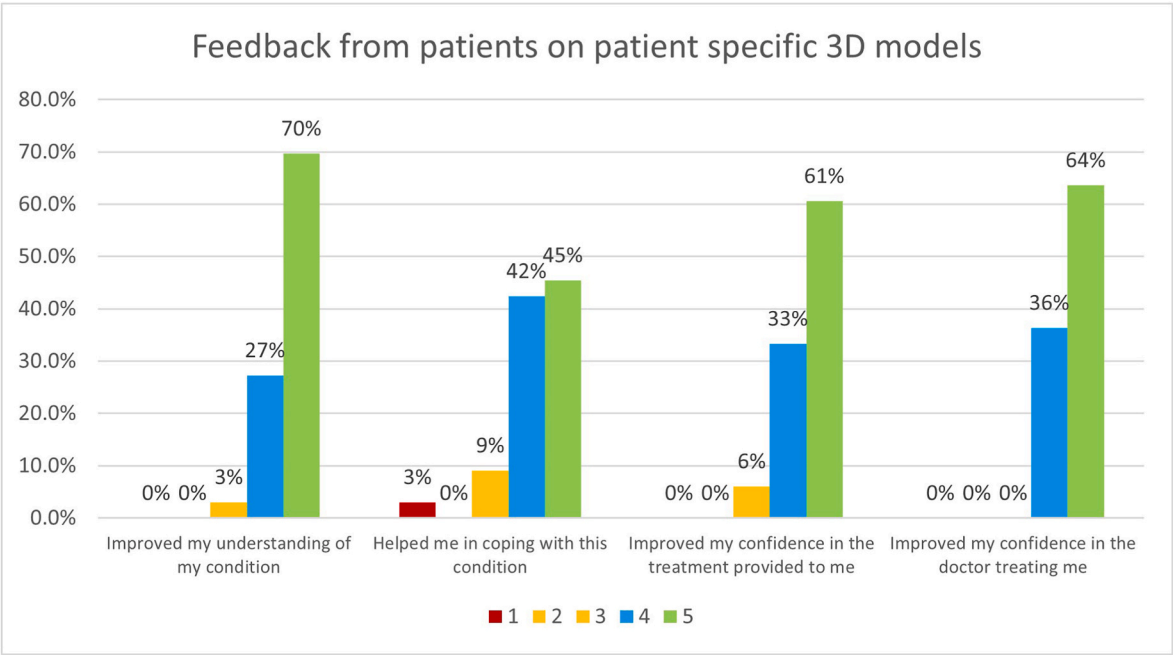


Fig. 3. Patients' rating on 4 different aspects of patient specific 3D models. Each column represents the percentage of patients indicating 1 to 5 for various aspects of the 3D models.

From the results, we clearly demonstrated that majority of patients who had a specific 3D printed model used for preoperative consultation agreed on its effectiveness in helping them to better understand their condition. Patients also reported to experience more confidence in the

treatment and healthcare professionals providing the treatment. In this discussion, we thoroughly analysed the role of patient specific 3D models in improving the quality of patient care and the influence of patient age and education level on the effectiveness of these models in



Fig. 4. Overall satisfaction score (Histogram). Y axis shows satisfaction scores from 0 to 10. X axis denotes percentage of patients for each satisfaction score.

Table 1

Spearman coefficients calculated at $p < 0.05$ for correlations between ratings for the 4 different areas, satisfaction level and patients' age. The results were all statistically insignificant.

	Spearman Coefficient	P value
Improve understanding of patient's condition	-.063	.364
Help patient in coping with the condition	-.187	.148
Improve patient's confidence in the treatment	.100	.289
Improve patient's confidence in the doctor	.000	.500
Overall satisfaction level	-.247	.083

Table 2

Mean ratings and satisfaction level of patients in pre-secondary and post-secondary groups.

	Highest education level		P value
	Pre-secondary	Post-secondary	
	Mean (SD)	Mean (SD)	
Sample size	17	16	
Improve understanding of patient's condition	4.59 (.618)	4.75 (.447)	.581
Help patient in coping with the condition	4.06 (1.03)	4.50 (.632)	.231
Improve patient's confidence in the treatment	4.41 (.618)	4.69 (.602)	.204
Improve patient's confidence in the doctor	4.53 (.514)	4.75 (.447)	.292
Overall satisfaction level	7.94 (1.48)	9.06 (.772)	.019

enhancing patients' experience.

4.1. Understanding

Our study showed that 97% patients indicated “strongly agree” and “agree” that 3D models had improved understanding of their conditions. Visual learning is one of the primary ways of acquiring information. 3D models of complex anatomical structures have become an important part of anatomy education for resident trainees and medical students as the model's allow easier visualisation of complicated anatomical pathologies.¹² Likewise, when these personalised 3D models are used in preoperative counselling, patients can visualise and subsequently comprehend the extent of injuries better. This is especially helpful as patients have different levels of health literacy and most of them do not have the relevant medical knowledge to derive a 3-dimensional

understanding from 2D CT scans. An added benefits of deepened understanding of traumatic injuries by patients is that they will be more compliant with post-operative rehabilitation, resulting in higher over quality of patient care.⁹ The subgroup analysis showed that age and education level do not influence perceived effectiveness of patient specific models in improving patients' understanding of their condition as there was also no significant correlation noted between patients' age and ratings on patient understanding and no significant differences between ratings on patient understanding in pre-secondary group and post-secondary group.

4.2. Coping

On a similar note, a high percentage of 87% patients agreed or strongly agreed that the customised models improved their self-efficacy in coping with their traumatic injuries. *Self-efficacy* refers to an individual's confidence in his or her ability to successfully carry out a specific task or behaviour.¹³ A negative but poor correlation was noted between age and ratings on boosting self-efficacy. Pre-secondary and post-secondary groups have comparable ratings on enhancing patients' self-efficacy. One main source of influence in the development of self-efficacy is physiological cues. Unpleasant physiological feelings such as expected bodily discomfort may make patients doubt their ability to cope with the injury and attain recovery. On the other hand, positive physiological cues such as pain relief provided by the treatment reinforce patients' confidence. Customised 3D models improved patients perceived self-efficacy by helping them to better understand the reason for their physical discomfort from the injury and anticipate the relief of pain and psychological stress through the treatment such as realigning displaced fracture with plates and screws as demonstrated by physicians on the models. However, a smaller than expected percentage of patients indicated strongly agree (45%) for the models' usefulness in boosting their self-efficacy in dealing with their condition compared to all other proposed benefits evaluated in the survey. This can be explained by the interplay of other factors that contribute to the construct of self-efficacy including verbal persuasion by physicians and previous experiences coping with orthopaedic trauma injuries.¹⁴ Patients who receive encouragement from their families and physicians will have higher level of self-efficacy and vice versa. A qualitative study could be conducted to determine the reasons for this lower percentage and other factors which can help boost self-efficacy level in patients to develop better patient care.

4.3. Confidence

Our study results have demonstrated that more than 90% of patients agree and strongly agree that customised 3D printed models have boosted their confidence in both the treatment and the doctor providing the treatment. Studies have shown that physicians' behaviour that inspire confidence and trust in doctors include thorough evaluation of problems, clear and complete communication, partnership building and the provision of effective treatment.¹⁵ We believe that using patient specific 3D models in preoperative counselling facilitates the delivery of clear and interactive communication as the models enable patients to directly visualise the complex anatomy and structural relationships, and doctors to subsequently demonstrate how the operation will be carried out on the model. According to a study on health literacy among lower limb trauma patients, half of the patients were unaware of the bone fractured and almost three quarters did not know what implants were used in the surgery.¹⁶ Patients may therefore perceive these 3D models as a gesture of invitation from the doctors to participate in the management of their conditions therefore facilitating partnership building between doctors and patients.

4.4. Satisfaction

A significant difference in overall satisfaction level on the use of patient specific models in preoperative counselling was noted between patients in pre-secondary (7.94) and post-secondary group (9.06). An Australian study showed that patients with higher education level perceived their participation in the decision-making process as a shared responsibility with the clinician whereas patients with lower education level perceived their participation as giving consent based on provided options.¹⁷ Patients with higher education levels are more willing to take the initiative to make sense of their condition and treatments available to reach shared responsibility with their doctors. They are therefore more likely to appreciate the use of customised models in preoperative consultation to help them develop clearer understanding of their conditions and treatment and higher level of involvement in healthcare decision-making which is associated with greater overall satisfaction.¹⁸

4.5. Limitations

One limitation of our study is its relatively small sample size (n = 26). The results may be underpowered and not be sufficiently representative of the advantages of these customised 3D models. Small sample size may lead to false positive results or over-estimation of the magnitude of association.¹⁹ For example, the weak positive correlation identified between the models' effectiveness in improving patients' understanding of their condition and boosting their confidence in the treatment is inconclusive as it is statistically insignificant.

5. Conclusion

To conclude, patient specific 3D models not only aid surgeons in preoperative planning but also largely improves the quality of doctor patient communication when incorporated into consultation. The models encourage patients' participation in their care planning through enhancing patients' understanding of their injury, promoting a patient-centred environment in the healthcare system. Age was not found to be a contributing factor in patients' perception of the 3D models, but we found that patients with higher education level experience higher overall satisfaction of patient care with the use of customised 3D models. 3D printing may be used as a useful tool to bridge the gap between patient and physician understanding and communication, working towards better value in healthcare. From this study, we recommend that healthcare professionals consider using 3D patient specific models in doctor-patient communication due to their potential benefits of improved understanding, confidence, and satisfaction in patients. However, the advantages of using customised 3D models in patient communication should be further validated by a more extensive study with greater sample size.

Funding sources

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Ethical statement

Ethical committee approval is not required for this study.

Patient consent

Patient consent was obtained verbally before questionnaires were

administered.

CRediT authorship contribution statement

Carol Xiaoshu Zhao: Formal analysis, Data curation, Writing – original draft, Visualization. **Michael Yam:** Conceptualization, Methodology, Validation, Investigation, Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

There is no conflict of interest to disclose.

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